Error-Gated Endpoint Evaluation of Michell Resistance for Low-Froude B-Spline Hulls

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Abstract

Reliable low-Froude wave-resistance prediction requires both resolving an increasingly oscillatory Michell integral and estimating its algebraic tail. Michelsen's 1960 dissertation and 1972 sequel developed analytic reductions of Michell resistance for polynomial and orthogonal-basis hull descriptions. We revive that lineage as a validated implementation for tensor-product B-spline hulls of degree at most 16 in each direction. Repeated integration by parts expresses the exact inner transform as endpoint waves. At low Froude number, an absolute bound permits the omission of terms below the waterline. Pair expansion then reduces resistance to kernels $K_s(\omega) = \int_1^\infty e^{i\omega\lambda}\lambda^{-s}(\lambda^2-1)^{-1/2}\,d\lambda$*, analytically continued Bickley functions. An exact contour rotation converts each sufficiently separated nonzero-frequency pair to a Gaussian-decaying integral whose quadrature cost does not grow with frequency. The reported error includes coefficient construction, endpoint accumulation, omitted submerged pairs, and coarse/fine contour quadrature; the dispatcher falls back to a real-axis marcher when the estimate exceeds the requested tolerance. For the standard Wigley hull at* $Fn = 0.02$*, the method uses 1,152 kernel nodes instead of 511,504 inner-transform evaluations, reduces median runtime from* 2.151×10^{1} ms *to* 3.100×10^{-2} ms *(about 700-fold), and differs from an independent real-axis reference by* 1.53×10^{-13} *relatively. The implementation is an engineering improvement for upright symmetric monohulls with coarse, well-separated active endpoints. Its contour-discretization estimate is not a formal interval certificate.*

KEY WORDS: Wave resistance; Michell integral; low Froude number; oscillatory quadrature; B-splines; Bickley–Naylor function.

Introduction

Michell's 1898 thin-ship theory expresses wave resistance as a one-dimensional integral of the squared Fourier–Laplace transform of the longitudinal hull slope [Michell, 1898, Tuck, 1989]. Compared with nonlinear free-surface methods, Michell's formula costs little and retains the bow–stern interference that couples resistance to speed and hull shape. Designers therefore use it for preliminary design, multihull studies, and mathematical shape optimization [Lazauskas, 2009, Dambrine et al., 2016]. At low speed, however, a fast result is useful only if its error estimate remains trustworthy.

An analytic reduction lineage addresses the same computational goal. Michelsen [1960], building on Birkhoff and Kotik's transformation, splits the calculation into a hull function and a speed-dependent Michell function, derives a convergent special-function series for polynomial hull functions, and proposes tabulation for systematic design. Michelsen [1972] later uses Gegenbauer source distributions and orthogonality to reduce the integral to a finite double sum. The verified abstract of de Sendagorta and Grases [1988] likewise describes rapidly convergent series, products of shape functions, and shape-independent velocity functions for tabulation and computer-aided design. We treat this history as the method's foundation and contribute an implementation-grade B-spline realization with measured error accounting and automatic fallback.

The apparent simplicity conceals a difficult numerical limit. Let $\nu = g/U^2$, where U is speed and g gravitational acceleration. As U and therefore $Fn = U/\sqrt{gL}$ decrease, the phase $\nu\lambda x$ oscillates increasingly rapidly over the unbounded outer coordinate λ. Tuck used Filon-like longitudinal integration, while Lazauskas used exact piecewise-quadratic formulas and a fixed angular rule; Lazauskas reports that very low Froude number remains exceptional [Tuck, 1989, Lazauskas, 2009]. A real-axis adaptive marcher must spend more effort as frequency rises and must decide where to truncate the oscillatory algebraic tail. At the pre-fix revision `64925d4`, one quiet-window criterion reported 5.123×10^{-9} relative error at $Fn = 0.05$, but the measured error was 5.781×10^{-7}, about 113 times larger. The frozen kernel advances quiet windows by oscillatory phase alone and adds the refinement and tail estimates. Across $0.02 \le Fn \le 1$, the corrected estimate covers measured error by about fourfold. In the four low-Froude cases reported below, measured error falls by factors of 3.2–4.5, and evaluations fall by 42–51%.

Earlier work also supplies the endpoint and contour ingredients. Wehausen [1973] surveys classical thin-ship theory and its limits. Keller and Ahluwalia show that bow and stern waterline data control the small-Froude wave field and resistance [Keller and Ahluwalia, 1976]. Gotman separates polynomial-hull contributions into endpoint derivatives and bow–stern interference terms [Gotman, 2002]. Huybrechs and Vandewalle develop numerical steepest descent [Huybrechs and Vandewalle, 2006], and Motygin applies it to the Kelvin Green-function integral in linear ship-wave theory [Motygin, 2017]. We combine these ideas in four steps:

1. decompose every polynomial B-spline span exactly into finitely many endpoint terms, including terms at interior and full-multiplicity chine knots;
2. bound every resistance pair that contains an omitted submerged endpoint;

3. identify the retained pair kernels as analytically continued Bickley–Naylor functions and evaluate them on a frequency-scaled Gaussian contour; and
4. accept the reduced result only when its error estimate meets the requested tolerance; otherwise use the general real-axis method.

These are implementation deltas within established analytic and numerical traditions, not claims of method priority. The accessible publisher and index records establish the statements above for Michelsen [1972] and de Sendagorta and Grases [1988]; their full texts remain due-diligence items.

Michell resistance and spline geometry

Let $y = f(x,z) \geq 0$ denote the half-breadth of an upright symmetric hull, with $z \geq 0$ measured downward from the undisturbed waterline. In the normalization used by the implementation,

$$R_\mathrm{w} = \frac{4\rho g^2}{\pi U^2}\int_1^\infty \frac{\lambda^2}{\sqrt{\lambda^2-1}}\,|F(\lambda)|^2\,d\lambda, \tag{1}$$

$$F(\lambda) = \iint_\Omega f_x(x,z)\mathrm{e}^{-\nu\lambda^2 z}\mathrm{e}^{\mathrm{i}\nu\lambda x}\,dx\,dz, \qquad \nu = \frac{g}{U^2}. \tag{2}$$

The hull is a clamped, non-rational tensor-product B-spline surface. On a knot rectangle $S = [x_0,x_1]\times[z_0,z_1]$, its longitudinal derivative is an exact bivariate polynomial,

$$f_x|_S = \sum_{a=0}^{p-1}\sum_{b=0}^{q} c_{ab}^S (x-x_0)^a (z-z_0)^b. \tag{3}$$

The method extracts these coefficients once from exact spline derivatives at span corners; it does not sample stations or waterlines. The implementation discards the zero-length intervals created by repeated knots. An interior knot whose multiplicity equals the degree represents a continuous chine exactly.

Exact endpoint decomposition

The finite polynomial degree makes repeated integration by parts terminate. For a polynomial p of degree m,

$$\int_{x_a}^{x_b} p(x)\mathrm{e}^{\mathrm{i}kx}\,dx = \sum_{r=0}^{m}\frac{(-1)^r}{(\mathrm{i}k)^{r+1}}\left[p^{(r)}(x)\mathrm{e}^{\mathrm{i}kx}\right]_{x_a}^{x_b}, \tag{4}$$

$$\int_{z_a}^{z_b} p(z)\mathrm{e}^{-\kappa z}\,dz = -\sum_{u=0}^{m}\frac{1}{\kappa^{u+1}}\left[p^{(u)}(z)\mathrm{e}^{-\kappa z}\right]_{z_a}^{z_b}. \tag{5}$$

Apply equations (4) and (5) to every monomial in equation (3), with $k = \nu\lambda$ and $\kappa = \nu\lambda^2$. The result is an exact endpoint representation.

Proposition 1 (Finite endpoint representation)**.** *For a polynomial tensor-product B-spline hull with longitudinal degree at least one, the transform in equation (2) can be written*

$$F(\lambda) = \sum_{j=1}^{M} c_j\lambda^{-n_j}\mathrm{e}^{\mathrm{i}\nu\lambda x_j}\mathrm{e}^{-\nu\lambda^2 z_j}, \qquad n_j \geq 3, \tag{6}$$

where (x_j, z_j) are endpoints of nonzero knot spans and the complex coefficients c_j are finite combinations of span-polynomial endpoint derivatives. Equal endpoint/power triples may be combined exactly.

Proof. Equations (4) and (5) replace a monomial of derivative orders (r,u) by endpoint factors proportional to

$$(\nu\lambda)^{-(r+1)}(\nu\lambda^2)^{-(u+1)}.$$

The resulting power of λ is $n = r + 2u + 3$. Both sums terminate at the polynomial degrees, and summing the finite contributions from all nonzero knot rectangles gives equation (6). □

Tests compare equation (6) directly with independent exact-moment calculations for both the Wigley hull and a multi-span hull with a full-multiplicity chine. Across $1 \leq \lambda \leq 20$, every discrepancy falls below the scaled 5×10^{-10} threshold.

Waterline reduction and omission bound

Partition equation (6) into waterline terms W, for which $z_j = 0$, and submerged terms S, for which $z_j > 0$. At low Froude number, every submerged term contains at least $\mathrm{e}^{-\nu z_j}$ because $\lambda \geq 1$. Retaining only W gives

$$\widetilde{F}(\lambda) = \sum_{j\in W} c_j\lambda^{-n_j}\mathrm{e}^{\mathrm{i}\nu\lambda x_j}. \tag{7}$$

Expanding $|\widetilde{F}|^2$ before outer integration removes the nonanalytic complex conjugation from the integrand:

$$\widetilde{R}_\mathrm{w} = C\sum_{i,j\in W}\Re\left\{c_i\overline{c_j}K_{s_{ij}}(\omega_{ij})\right\}, \qquad C = \frac{4\rho g^2}{\pi U^2}, \tag{8}$$

where $s_{ij} = n_i + n_j - 2 \geq 4$, $\omega_{ij} = \nu(x_i - x_j)$, and

$$K_s(\omega) = \int_1^\infty \frac{\mathrm{e}^{\mathrm{i}\omega\lambda}}{\lambda^s\sqrt{\lambda^2-1}}\,d\lambda. \tag{9}$$

Proposition 2 (Submerged-pair error bound)**.** *Let $\mathcal{P}_S = \{(i,j) : i \in S \text{ or } j \in S\}$. The resistance discarded by replacing F with $\widetilde{F}$ satisfies*

$$\left|R_\mathrm{w} - \widetilde{R}_\mathrm{w}\right| \leq C\sum_{(i,j)\in\mathcal{P}_S}|c_i c_j|\mathrm{e}^{-\nu(z_i+z_j)}K_{s_{ij}}(0). \tag{10}$$

Proof. Expand $|F|^2 - |\widetilde{F}|^2$ into precisely the ordered pairs in $\mathcal{P}_S$. Apply the triangle inequality, use $|\mathrm{e}^{\mathrm{i}\omega\lambda}| = 1$, and bound $\mathrm{e}^{-\nu\lambda^2(z_i+z_j)} \leq \mathrm{e}^{-\nu(z_i+z_j)}$ for $\lambda \geq 1$. The remaining positive integral is $K_{s_{ij}}(0)$. □

The zero-frequency kernel is analytic:

$$K_s(0) = \int_0^\infty \cosh^{-s} t\,dt = \frac{\sqrt{\pi}\,\Gamma(s/2)}{2\Gamma((s+1)/2)}, \tag{11}$$

A stable even/odd recurrence evaluates this kernel.

Bickley kernels and contour quadrature

With $\lambda = \cosh t$, equation (9) becomes

$$K_s(\omega) = \int_0^\infty \mathrm{e}^{\mathrm{i}\omega\cosh t}\cosh^{-s} t\, dt = \mathrm{Ki}_s(-\mathrm{i}\omega), \tag{12}$$

where the final equality denotes analytic continuation of the Bickley–Naylor function [Bickley and Nayler, 1935, NIST Digital Library of Mathematical Functions, 2026].[1] This identity supplies recurrences and other possible implementations. The present solver instead evaluates the contour directly.

The substitution $\lambda = 1 + t^2$ removes the square-root endpoint in equation (9):

$$K_s(\omega) = 2\mathrm{e}^{\mathrm{i}\omega}\int_0^\infty \frac{\mathrm{e}^{\mathrm{i}\omega t^2}}{(1+t^2)^s\sqrt{2+t^2}}\, dt. \tag{13}$$

For $\omega > 0$, analyticity in the intervening sector permits the exact rotation $t = \mathrm{e}^{\mathrm{i}\pi/4} y/\sqrt{\omega}$, giving

$$K_s(\omega) = \frac{2\mathrm{e}^{\mathrm{i}(\omega+\pi/4)}}{\sqrt{\omega}}\int_0^\infty \frac{\mathrm{e}^{-y^2}\, dy}{(1+\mathrm{i}y^2/\omega)^s\sqrt{2+\mathrm{i}y^2/\omega}}. \tag{14}$$

Conjugation gives the negative-frequency kernels. The rotation replaces oscillation with Gaussian decay and leaves slowly varying frequency dependence in the amplitude. The same numerical-asymptotic principle yields frequency-independent work in high-frequency scattering [Gibbs et al., 2020]; here, the quadratic phase gives the contour in closed form.

For completeness, take the principal square root in equation (13). The integrand's poles $t = \pm\mathrm{i}$ and branch points $t = \pm\mathrm{i}\sqrt{2}$ lie outside the closed sector $0 \le \arg t \le \pi/4$. It is therefore analytic throughout the deformation. On a closing arc $t = R\mathrm{e}^{\mathrm{i}\phi}$, $|\mathrm{e}^{\mathrm{i}\omega t^2}| = \mathrm{e}^{-\omega R^2\sin(2\phi)} \le 1$, while the algebraic factor including dt is $O(R^{-2s})$. The arc contribution vanishes as $R \to \infty$; thus the rotation is valid for every kernel order used here, $s \ge 4$.

The implementation normally applies 24- and 48-point Gauss–Legendre rules on $0 \le y \le 8$. When $s/|\omega| \ge 2$, the continued algebraic factor is narrow near the origin, so the code uses 48- and 96-point rules. The neglected Gaussian tail obeys

$$|K_s - K_s^{(8)}| \le \frac{\mathrm{e}^{-64}}{8\sqrt{2\omega}}, \qquad \omega > 0, \tag{15}$$

using $|1 + \mathrm{i}y^2/\omega| \ge 1$, $|\sqrt{2+\mathrm{i}y^2/\omega}| \ge \sqrt{2}$, and the standard Gaussian-tail bound. At the minimum supported nonzero frequency $\omega = 25$, this is below 4×10^{-30}. The selected coarse/fine difference estimates discretization error. This difference remains empirical. Thus the submerged-term bound and analytic contour-tail bound are rigorous, but the total floating-point error is not certified.

For fixed geometry, a nonzero endpoint pair costs either 72 or 144 scalar nodes; increasing frequency never increases that cost. Self pairs use equation (11) without quadrature. The biquadratic Wigley representation has one longitudinal span and one vertical span. Its 16 endpoint terms include eight at the waterline, which form 16 distinct nonzero-frequency bow–stern pairs. Each uses the 24/48 rule, for $16 \times 72 = 1{,}152$ scalar nodes in all; equal-position pairs use equation (11). The committed harness checks each count.

Hybrid algorithm and scope

The reduced method supplements the general solver. The following gates choose between them:

1. validate the physical conditions and exact spline geometry;
2. form and combine the exact endpoint terms in equation (6);
3. reject the specialization unless the hull is upright, symmetric, has its spline domain beginning exactly at $z = 0$, and has longitudinal degree at least one;
4. reject it if any distinct retained endpoint pair has $0 < |\omega_{ij}| < 25$;
5. evaluate equation (8), the bound equation (10), and the weighted coarse/fine contour difference; and
6. use the reduced result if their combined relative estimate meets the requested tolerance; otherwise run the general real-axis marcher.

The frequency gate reserves the fixed contour for sufficiently rapid oscillation. For the Wigley bow–stern pair, $\omega = \nu L = gL/U^2 = 1/Fn^2$, which equals the implementation's $\nu|x_i - x_j|$. Thus $\omega \ge 25$ corresponds to $Fn \le 0.2$. The same condition makes the geometry restriction explicit. For m equal longitudinal spans whose adjacent endpoints remain active, $\omega_{\mathrm{adj}} = 1/(mFn^2)$, so the separation gate requires

$$m \le \frac{1}{25Fn^2}. \tag{16}$$

At $Fn = 0.05$, for example, this permits at most 16 such spans. A retained multi-span endpoint map with 360 distinct nonzero-frequency pairs would use $360 \times 72 = 25{,}920$ ordinary-rule nodes, compared with 1,152 for the 16-pair Wigley map; this is node-count arithmetic, not a runtime benchmark. One closer endpoint invalidates the specialized path for the whole hull. The error gate then bases the final choice on the actual hull, speed, and resistance cancellation rather than on nominal Froude number alone. At the default relative tolerance 10^{-5}, it rejects the Wigley reduction at $Fn = 0.08$ and accepts it at $Fn = 0.05$.

The implementation uses safe Rust and no external numerical library. Tests check the reduced solver at three independent levels: exact inner decomposition, isolated Bickley kernels, and total resistance. The broader validation harness checks Froude similarity, zero camber for symmetric sides, classical catamaran $4\cos^2$ interference, tolerance self-convergence, and reported against observed error.

[1]The 1935 paper is by W. G. Bickley and J. Nayler; later literature and DLMF use the conventional function name "Bickley–Naylor." We retain each spelling in its historical context.

Numerical experiments

Independent reference

The standard Wigley half-breadth is

$$f(x,z) = \frac{B}{2}\left(1 - \frac{4x^2}{L^2}\right)\left(1 - \frac{z^2}{T^2}\right), \quad |x| \le L/2, \quad 0 \le z \le T, \tag{17}$$

with $L = 1.000 \times 10^1$ m, $B = 1.000$ m, and $T = 6.250 \times 10^{-1}$ m. The independent reference shares neither the production moment code nor its outer integrator. Instead, it evaluates the analytic Wigley transform on the positive real λ axis with 16-point Gauss–Legendre panels and Kahan-compensated accumulation. Panel boundaries advance the bow phase by at most $\pi/2$, and integration continues to $\lambda = 4000$. At $Fn = 0.02$, doubling this cutoff to 8000 changes the reference by 1.23×10^{-15} relatively. Because floating-point effects remain, this check measures resolution but does not provide an interval bound.

As a published-value anchor, Doctors and Beck [1987, Table 1] report the classical thin-ship value $10^3 C_w = 1.2486$ for this Wigley hull at $Fn = 0.35$. The committed harness gives $10^3 C_w = 1.247922$, a relative difference of 5.43×10^{-4} (0.0543%). We classify this agreement as a known result reproduced (class 1).

Table 1 shows two regimes. At $Fn = 0.08$, submerged terms remain material, and the estimate rejects the reduction at default tolerance. At $Fn \le 0.05$, reduced/reference differences range from 1.53×10^{-13} to 1.03×10^{-12} and remain within the reduced solver's empirical estimates. The general marcher's error, by contrast, grows as Froude number falls. At $Fn = 0.05$, its corrected estimate is 5.774×10^{-7}, versus 1.493×10^{-7} measured. The estimate therefore covers the observed error. This independently checked marcher provides every comparison in table 1.

The isolated kernel tests compare equation (14) against dense real-axis integration for $s \in \{4, 7, 10, 50, 98, 128\}$ and $|\omega| \in \{25, 100, 400\}$, with scaled discrepancy below 10^{-9}. The public degree envelope reaches at most $s = 98$; order 128 provides margin. The full Rust and Python-interface suites pass.

Runtime

Timing used an optimized build, 30 samples per case, and the default relative tolerance 10^{-5}. Each case received its own unmeasured warm call. The two method comparisons alternated execution order on every sample; other cases were sampled separately in the printed order. Absolute timings are host-dependent; the work counts and checksums are deterministic. The host was an Apple M5 MacBook Air running Darwin 25.5.0, with Rust/Cargo 1.96.0.

At $Fn = 0.02$, the frozen paired run gives about a 700-fold median speedup. Two immediately repeated paired runs gave 702–707-fold; an earlier cold-host run gave 608-fold, while the former fixed-order reviewer protocol gave approximately 750–865-fold. We therefore do not treat any point ratio as portable. Cost changes little between $Fn = 0.05$ and 0.02, as equation (14) predicts. During the 21-speed design sweep, the hybrid selects the path that satisfies the error gate. The checksum is $2.553251156101 \times 10^5$; it differs from the pre-fix value because the corrected marcher retains a positive tail that the earlier rule truncated.

Analytic lineage and implementation scope

The present solver revives a substantial analytic lineage. The equation-level comparison in table 3 separates the historical reductions from the implementation work reported here. For the 1972 and 1988 papers, the table states only what their verified publisher or index records establish; both full texts remain due-diligence items rather than dependencies of our classification.

The implementation differs from that lineage in five engineering details:

1. it operates span by span on tensor-product B-splines, including full-multiplicity chine knots;

2. it restricts both spline degrees to the independently validated range 1–16 and charges coefficient construction and endpoint cancellation;

3. it bounds the resistance contribution of every omitted submerged endpoint pair;

4. it evaluates retained K_s pairs on a frequency-scaled contour with a bounded, order-aware node count; and

5. it accepts the reduction only when the combined estimate meets the requested tolerance, otherwise returning to the general marcher.

These deltas are engineering improvements, not priority claims. A full method-by-method benchmark against the Michelsen and Sendagorta–Grases representations lies outside this paper and remains future work.

The endpoint structure is also established independently. Keller and Ahluwalia [1976] express leading small-Froude resistance and waves through bow and stern waterline slopes. Gotman [2002] repeatedly differentiates polynomial hull functions, separates endpoint self terms from bow–stern interference, and discusses higher endpoint derivatives. Our finite B-spline decomposition puts that mechanism into implementation form; it does not introduce a new physical asymptotic principle.

Tuck and Lazauskas also establish exact or Filon-like integration of piecewise-polynomial inner data [Tuck, 1989, Lazauskas, 2009]. As Lazauskas [2009] describes it, Michlet combines exact piecewise-quadratic longitudinal formulas with a fixed angular rule. Our solver retains the same thin-ship physics but, for accepted B-spline hulls, keeps low-Froude outer work from growing with frequency. We classify that result as an engineering improvement and make no head-to-head performance claim against the Michlet executable. Exact endpoint reduction also survives the outer modulus square: pair expansion converts it into a small set of reusable special-function kernels. Ruffa and Toni give finite Bessel–Struve module representations for integer-order Bickley functions [Ruffa and Toni, 2026]; their stability on the imaginary axis remains to be tested as a possible kernel backend.

Contour deformation is well established for oscillatory quadrature [Huybrechs and Vandewalle, 2006]. Motygin [2017] applies steepest descent and Clenshaw–Curtis quadrature in ship-wave theory, but to the oscillatory part of the Kelvin Green function rather than Michell's resistance integral. Automated methods now

Table 1: Resistance validation against the independent analytic-Wigley, resolved-real-axis reference. "Reduced estimate" combines the rigorous submerged-pair bound and empirical contour difference. The marcher uses requested rel_tol = 10^{-8} and at most six refinements.

Fn	Reference R_{w} (N)	Reduced rel. diff.	Reduced estimate	Marcher rel. diff.
0.08	1.946×10^{-1}	1.172×10^{-5}	3.814×10^{-5}	6.918×10^{-8}
0.05	1.058×10^{-2}	2.070×10^{-13}	3.728×10^{-12}	1.493×10^{-7}
0.03	5.114×10^{-4}	1.034×10^{-12}	3.462×10^{-12}	3.322×10^{-7}
0.02	4.417×10^{-5}	1.526×10^{-13}	7.620×10^{-13}	6.334×10^{-7}

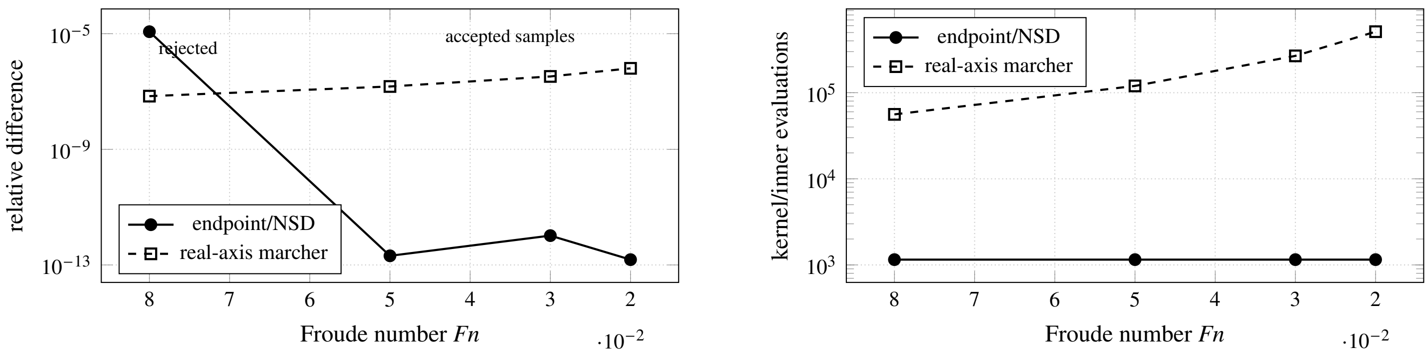

Figure 1: Observed accuracy (left) and work (right) as Froude number decreases. The independent reference resolves endpoint/NSD differences at the 10^{-13}–10^{-12} level, while geometry fixes its quadrature work. At default tolerance, the hybrid rejects the reduced result at $Fn = 0.08$ and accepts it at $Fn = 0.05, 0.03, 0.02$.

handle general phases and coalescing saddles [Gibbs et al., 2024]. Such methods could treat multihull transverse phases; the present quadratic phase needs only the closed contour in equation (14).

Limitations and open extensions

The evidence covers a narrow model: upright symmetric monohulls in linear, deep-water thin-ship theory.

- **Geometry.** The reduced solver supports upright symmetric single hulls of degree 1–16 in each spline direction. Its active endpoints must satisfy $\nu|x_i - x_j| \geq 25$, so it targets coarse, well-separated endpoint maps. Asymmetric, heeled, multihull, and closely spaced cases fall back to the general method.

- **Error certification.** The submerged-pair omission bound and Gaussian tail bound are analytic. The selected coarse/fine quadrature difference and floating-point roundoff are not interval-certified.

- **Reference floor.** The independent real-axis reference uses compensated summation through $\lambda = 4000$; at $Fn = 0.02$, a cutoff-doubling check changes it by 1.23×10^{-15}. Differences near the observed 10^{-13} scale are resolution measurements, not interval-certified digits.

- **Physical model.** Michell theory is linear, inviscid, slender, deep-water, and fixed-attitude. At very low speed, viscous effects may dominate the already small wave resistance. For free-surface flow past singular obstacles, low-Froude waves can be exponentially small, beyond all algebraic orders, and controlled by Stokes phenomena; that different problem is not resolved by better quadrature [Trinh and Chapman, 2015].

The highest-leverage numerical extension is multihull support. A transverse placement introduces phase $\nu y \lambda \sqrt{\lambda^2 - 1}$ and moving saddles that equation (14) does not contain. A uniform or automated contour must account for them. Retaining shallow submerged endpoints likewise produces mixed linear and quadratic complex phases. Further work should certify the contour error, test imaginary-argument Bickley recurrences, derive finite-depth kernels, and differentiate the endpoint representation for low-Froude hull optimization.

Conclusions

Finite integration by parts converts the B-spline inner transform into endpoint waves. Pair expansion then turns the outer modulus square into continued Bickley kernels, and contour rotation replaces real-axis oscillation with Gaussian decay. Geometry, rather than Froude number, sets the reduced quadrature work. The submerged-pair bound lets the hybrid reject the reduction whenever it exceeds the requested tolerance.

For the standard Wigley hull at $Fn = 0.02$, the reduced solver is more accurate than the general marcher and about 700 times faster in the frozen paired run. This result is an engineering improvement for the supported geometry, built from the analytic reduction lineage summarized in table 3. Claims beyond the current scope require broader geometry and certified total error control.

Table 2: Final release benchmark on the development host.

Case	Median (ms)	IQR (ms)	Diagnostic
21 speeds, $Fn = 0.10 \ldots 0.50$	13.035	12.996–13.064	corrected checksum
Default API, $Fn = 0.05$	0.030	0.030–0.030	1,152 nodes
Real-axis marcher, $Fn = 0.02$	21.506	21.464–21.557	511,504 evaluations
Endpoint/NSD, $Fn = 0.02$	0.031	0.030–0.031	1,152 nodes

Table 3: Equation-level comparison of polynomial and basis reductions of Michell resistance. "Record" denotes a verified publisher or bibliographic record where the full text was unavailable.

Source	Basis and reduction	Kernel or speed dependence	Error control and dispatch	Relation to this implementation
Birkhoff–Kotik via Wehausen [1973, eqs. 39–45]	Hull autocorrelation $M(u, v)$, with equivalent (P, Q) transforms and separable elementary ships	Geometry-independent kernel $K(\nu u, \nu v)$; a Y_0 reduction is also reported	Absolute convergence justifies the change of integration order	Establishes separation of hull data from a reusable kernel
Michelsen [1960, eqs. 3.3–3.4]	Polynomial hull function $H(\xi, \zeta)$ after the Birkhoff–Kotik transformation	Michell function $C(s, t)$ contains speed dependence	Convergence conditions accompany the transformation	Direct historical antecedent of shape/kernel separation
Michelsen [1960, eqs. 3.21–3.22, 3.38–3.40]	Monomial expansion of H; each coefficient contributes through a closed series expression	Confluent-hypergeometric, Bessel, and Struve functions; speed-and-degree terms can be tabulated	Appendix proves series convergence; computer tabulation is proposed	Antecedent of analytic polynomial reduction for design use
Michelsen [1972], verified record	Gegenbauer expansion of the center-plane Havelock-source distribution	Orthogonality reduces resistance to a finite double sum depending on source coefficients, Fn, and length–draft ratio	The accessible abstract states the reduction and identifies its governing inputs	Published JSR continuation of the basis-reduction program
de Sendagorta and Grases [1988], verified record	Products of integral shape functions	Rapidly convergent series with linear, shape-independent velocity functions suitable for tabulation	The abstract reports that few terms give suitable accuracy; detailed controls await source access	Antecedent of separated, table-driven computer-aided design
Present implementation	Degree-$\leq$ 16 tensor-product B-spline span polynomials reduced to active endpoint waves	Continued Bickley pair kernels on a frequency-scaled Gaussian contour	Coefficient, accumulation, omission, and quadrature estimates; explicit acceptance gate and real-axis fallback	Modern error-accounted realization for coarse, well-separated endpoints

Reproducibility statement

The implementation, tests, benchmark harness, generated results, and number audit are available in the public `michell` repository. On acceptance for submission, the immutable source, test, and artifact package will be archived and the placeholder `10.5281/zenodo.REPLACE-ME` replaced with its minted DOI. The current numerical kernel will be identified by the immutable local tag `paper-a-jsr-v2`; that tag will not be moved after creation.

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